DESIGN THINKING

Guide for research

Guilherme Victorino
Joana Mendonça
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1- INSPIRATION

- Understanding the challenge

In this phase, the aim is to get to the bottom of the initial problem through various sources and methods, including, for example, interviews and documentary research.

The aim is to get to the bottom of the initial problem using various sources, interviews and observation and empathy tools to achieve insights.

What we identified about the challenge

What we know

What we want to know more about

Rewriting the challenge

I- INSPIRATION - PREPARE RESEARCH

To gather insights and observe, we use a combination of different methods: collecting ethnographic data in the field, conducting interviews, reviewing patents, collecting data and trends, manufacturing processes, vendors and subcontractors. This information can be collected in notes, pictures, images and videos and recorded conversations

- Methods to be used:
  Ethnography; mobile ethnography, netnography
  Documentary research
  User observation
  Identifying and using extreme users
  Interviews

  We use observation and research to identify surprises and events that we don't expect, so we don't observe average users, but extreme cases, since extreme users have a more pronounced behavior.
  This observation work allows us to identify differences between what people say and what people do and think. At this stage, we also look for adaptations and improvisations that people may make and hidden passions.

Human Data and Models	Capture of Needs, Desires and Meanings	Simulation of Poss
- Anthropometric data sets and models	Verbally based	- Focus groups
- Biomechanical data sets and models		- Lead user design
- Psychophysical data sets and models	- Ethnographic interviews	- Co-design
- Cognitive data sets and models	- Questionnaires	- Storyboard futures
- Emotional data sets and models	- Day-in-the-life analysis	- Experience prototy
- Psychological data sets and models	- Activity analysis	- Para-functional pro
- Sociological data sets and models	- Cognitive task analysis	- Role playing
- Philosophical data sets and models	- The five whys	- Real fictions
	- Conceptual landscape	
	- Word-concept association	
	- Think aloud analysis	
	- Metaphor elicitation	
	- Be your customer	
	- Customer journey	
	- Extreme Users	
	- Personas	
	- Scenarios	
	- Brainstorming	
	- Contextual Inquiry	
	Non Verbally based	
	- Game playing	
	- Cultural Probes	
	- Visual journals	
	- Error analysis	
	- Fly-on-the-wall observation	
	- Customer Shadowing	
	- Body language analysis	
	- Facial coding analysis	
	- Physiological measures	
	- Electroencephalograms	

Table 1) Frequently deployed human centred design tools.

Fonte: Joseph Giacomin

TECHNIQUES

- Parallel worlds – other contexts from which we can make parallels, and adapt to our context
- Extreme users - allows you to find inconsistencies, their behavior is more prominent and therefore easier to identify. The hard part is sometimes finding the extreme users (the ideal is to observe the two ends of the spectrum in pairs).
- A day in the life - ethnographic technique - I ask someone to describe their day to me, with photos and comments, recording their frustrations or the things that don't work.
- Fly on the wall - passive observation
- Empathic experience - we impersonate the client
- Ways of obtaining information: Verbal (interviews); non-verbal (observation)

FERRAMENTAS: STAKEHOLDER MAP

- For this activity you will create a stakeholder map and then reflect on that process. To build the stakeholder map:

1. Summarize in a few sentences the project you're working on with this group of stakeholders.
2. Brainstorm an initial list of stakeholders.
3. Once you've brainstormed the list, think about how they're related, knowing that you can rearrange them in the future. Group the stakeholders by their relationship.
4. Think about potential categories for each related group, and work to name each category.
5. Ascribe motivations to each group of stakeholders - what is driving their behavior that is influencing you or your project? What's driving their decision-making behavior? Frame this in a positive way.
6. If possible, draw connections that reflect how different clusters might be related.

- Once you've created your map, complete this reflection:
- How did this process help you build your team?
- What stakeholders did you include that you may not have originally included?
- What are the advantages to building stakeholder maps as part of the innovation process?
- Describe a feasible method for you to iterate your stakeholder map as you move through a project.

1- INSPIRATION - PREPARE RESEARCH

- You can use different methods to go into depth on different issues, as in the example table - make a table to prepare your work and decide which group member does what

How can we help the individual student to find the right education according to his/hers preferences?			
Concept/topic	**Technique**	**Tool**	**Responsible**
Difference between a student that made a successful decision and a bad one.	- Extreme user - Desktop research	Interviews	Jesper
Students changing degrees. Why	- Interview for empathy - Desktop research	Interviews	Davis
Validate “Common factors which may lead to unsatisfactory decisions” found in Phase 1. Look for more factors.	- Interview for empathy - Desktop research	Interviews	Andreas
How criteria’s priority changes before /after applying university?	- Assume Beginner’s Mindset - Desktop research	Interviews	João
How do young people make decisions in a general context?	- Parallel Universe	Observation (where to have lunch, what parties to enter, āwhat to do after school, shopping behaviour etc.)	All

Prepare Research

Identifying Sources of Inspiration

Who are all the people involved in your topic? Who can represent extreme behaviors related to your topic? Which experts do you want to meet to find out more about your topic? List the candidates you think will give you the most inspiration and circle 3-5 with whom you want to engage first.

People

Identifying Places of Inspiration

Where can you go to have an inspiring experience related to your challenge? What are analogous scenarios or extreme experiences where you can witness similar or relevant behaviors and activities in a different context? List as many places as you can and circle 3-5 that you are most excited to observe first.

Places

Selecting research participants

Who specifically do you want to talk to and learn from? Create detailed descriptions for at least 3 different users or sources of inspiration. Make sure you cover a range of gender, experience, ethnicity, etc.

Person:

Description:

Person:

Description:

Person :

Description:

Person :

Description:

Building an Interview Script

What do you want to learn in order to better understand the challenge at hand? What do you hope to understand about people's motivations and frustrations? What do you want to learn about their activities?

Make a file for each person

Interviewee:_____

What are some specific questions you can ask to open up the conversation?

What are some questions that can help you begin to understand this person's hopes, fears and ambitions?

What are some ways to deepen the conversation, to find out even more about this person's perspective?

Building an observation guide

What do you want to learn from this observation? Capture themes and questions that you want to make sure you get to when you visit the site.

Make a file for each place

Place:_____

What are some things you want to make sure you observe while visiting this place?

What are some things you can do to gain inspiration in this place?

Inspiration

Capture what you see and hear during interviews and observations. Capture concrete quotes. Separate your observations from your interpretations. Look for solutions and adaptations that people have made to make a system that better suits their needs, such as books stacked under a laptop to make the screen a better viewing height.

Make a file for each place and interview

Place/Interview:_____

Observations and quotes	Interpretation

2- INTERPRETATION

- Interpreting the results and discovering

In the inspiration phase, we understand the challenge and gain empathy.

In this phase, we want to interpret what we have collected, give it meaning and define insights from our observations and interviews.

Telling the story

Immediately after the interview or observation, record what you have learned. Record an observation, highlight from the story or quote.

Place/interview:_____

What was the most memorable and surprising story?

What was interesting about your interaction with the environment?

What is the person's greatest motivation? And their biggest concern?

What's your biggest frustration?

What questions would you like to ask?

Share the most inspiring stories

Share the stories as a group and capture the most interesting observations and learnings on notes or post-it notes (physical or virtual).

What I learned from the team

What I learned from the team

What I learned from the team

What I learned from the team

What I learned from the team

Find themes

Look for themes, patterns and connections in your notes. Gather your notes around themes or patterns that you identify. Create titles for each category that capture these themes

Themes

Making Sense of Discoveries

Take a closer look at the themes and find overlaps, patterns and tensions that relate to each other. Can you group several related themes into larger categories?

Categories

Did you find any contradictions? Any unexpected learning or anything surprising? Why?

Unexpected and surprising

Which categories do you want to explore?

To explore

Defining Insights

Insights are a concise expression of what you have learned from your research and inspiration. They are the "aha" moments and unexpected learnings.

Insights are questions that invite us to explore

Insight

Insight

Insight

Insight

Making Insights Actionable

Actionable ideas start with the right question, which identifies important opportunities through great INSIGHTS. Each "How Can We" (HMW) question will only address one part of your challenge so you can create multiple HMWs and prioritize them for ideation.

How Might We

Avoid questions that immediately imply a solution

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